

★ WEEK 6 · PART 2 · PRACTICE

Granger Causality, Worked by Hand

Does One Series Help Forecast Another - and Which Way?

The F-test on cross-lags, and the four outcomes

Part 2's test, with the arithmetic shown. We run the Granger F-test - does adding X's past lags cut the forecast error for Y? - then test the reverse direction to classify the pair, and confront why a significant result is prediction, not proof of cause.

2P

PART 2 · APPLICATIONS

Prediction, Not Proof

Case A: the Granger F-test - fit Y on its own past, then add X's lags, and test whether the error drop is significant.

Case B: test the reverse direction to place the pair among the four outcomes, and see why Granger causality isn't real causality. Then a Claude Code lab.

What Part 2 Told Us

Now we test it - with numbers

1

Granger causality is about forecasts

X Granger-causes Y if X's past improves the forecast of Y beyond Y's own past - a statement about prediction, not mechanism.

2

It's an F-test on cross-lag coefficients

Fit Y restricted, then unrestricted with X's lags added. If the residual sum of squares drops enough, X helps.

3

A significant result is not real causality

Prediction can come from a common driver or anticipation. Granger causality answers a narrower, forecasting question.

Everything here is a calculator exercise: one F-test, one reverse test, one classification.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a restricted and unrestricted fit.
- Compute the F, or test the reverse direction.
- Then read the verdict - carefully.

What you should watch

- How the RSS drop becomes an F-statistic.
- How the reverse test sets the direction.
- Why prediction is not the same as cause.

The Granger F-Test

Does X help forecast Y?

We ask whether a sector index's past returns help forecast a stock's return, beyond the stock's own history. Fit the stock on its own lags - that's the restricted model - then add the sector's lags for the unrestricted one. If the residual error drops enough, an F-test declares the sector Granger-causes the stock. We compute it by

The setup

- Y is the stock, X the sector index.
- Two lags each, 100 observations.
- Restricted RSS 52, unrestricted 45.

What we'll do by hand

- Form the F from the RSS drop.
- Compare to the F critical value.
- Decide whether X helps.

Restricted vs Unrestricted

The forecast error, with and without X

$$\text{restricted: } Y_t = \sum a_i Y_{t-i} \quad \text{vs} \quad \text{unrestricted: } Y_t = \sum a_i Y_{t-i} + \sum b_i X_{t-i} \quad (\text{test } b_i=0)$$

The two models

- Restricted: Y on its own past only.
- Unrestricted: add X's past lags.
- The test asks if those lags matter.

The null

- All of X's lag coefficients are zero.
- Then X adds nothing to the forecast.
- The F-test tries to reject that.

The F-Statistic

From the drop in residual error

$$F = \frac{(\text{RSS}_r - \text{RSS}_u)/q}{\text{RSS}_u/(T-k)} = \frac{(52-45)/2}{45/95} = \frac{3.5}{0.474} = \mathbf{7.39}$$

The numerator

- The RSS falls from 52 to 45.
- A drop of 7, over 2 restrictions.
- That's 3.5 per restriction.

The denominator

- The unrestricted RSS over its degrees of freedom.
- 45 over 95, about 0.47.
- The ratio is an F of 7.39.

The Verdict

Against the F critical value

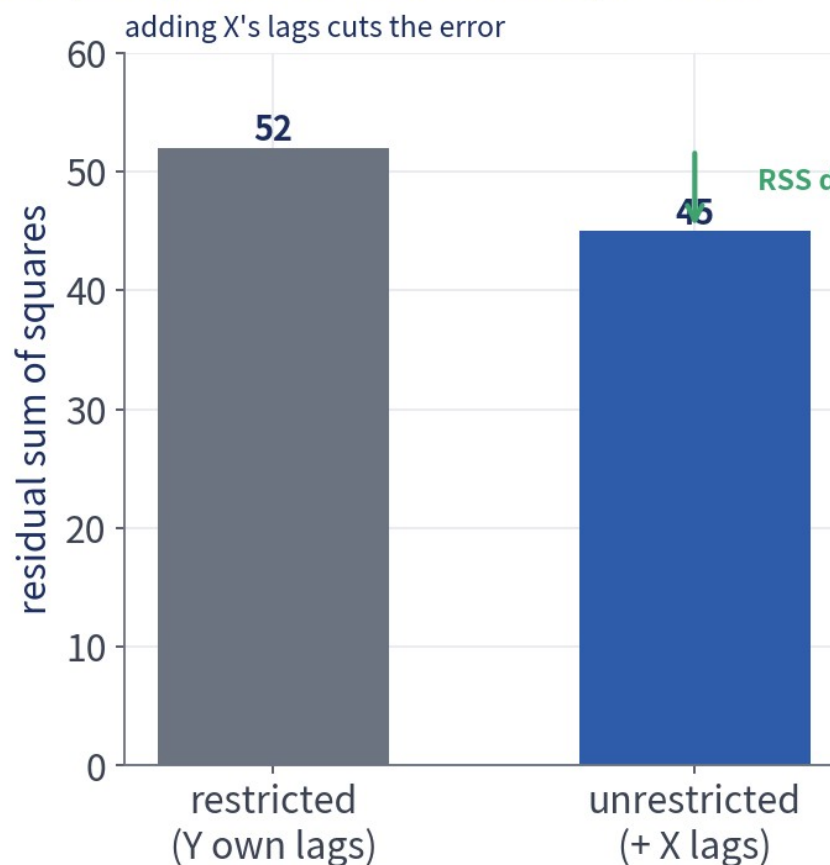
$$7.39 > \underbrace{3.09}_{F(2,95) \ 5\%} \Rightarrow \text{reject } b_i=0 \Rightarrow X \text{ Granger-causes } Y \text{ (its past adds forecast power)}$$

The F of 7.39 far exceeds the 5% critical value of 3.09, so we reject the null that X's lags are all zero. The sector's past does improve the stock's forecast - X Granger-causes Y.

The Error Drop, and the F

Adding X's lags cuts the residual sum of squares from 52 to 45 - a drop of 7. Scaled into an F-statistic, that gain reaches 7.39, well above the 3.09 bar. The reduction is large enough that random chance is an implausible explanation. X's past genuinely helps

Case A: X's past lags cut the forecast error enough to matter



What 'Helps Forecast' Means

A prediction claim, precisely

What it does say

- X's past cuts Y's forecast error.
- The reduction is statistically real.
- X carries predictive information about Y.

What it doesn't say

- That X mechanically drives Y.
- That acting on X will change Y.
- Only that X predicts Y - for now.

Direction, and the Caveat

Which way, and what it really means

Granger causality has a direction, so we test the reverse: does the stock's past help forecast the sector? Running the same F-test the other way gives a small, insignificant statistic - so the relationship is one-directional. That places the pair in one of four outcomes. But even a strong result is prediction, not proof of cause

The reverse test

- Now forecast X from its own past.
- Add Y's lags and test them.
- The RSS barely moves.

What we'll do by hand

- Compute the reverse F.
- Classify among the four outcomes.
- Then weigh the causality caveat.

The Four Outcomes

Every pair falls into one

four outcomes: $X \rightarrow Y$ only | $Y \rightarrow X$ only | both (feedback) | neither

How to place a pair

- Test both directions with the F.
- Each direction is significant or not.
- The two answers pick the outcome.

The four cells

- One-way each direction, feedback, or none.
- Feedback means both tests reject.
- Neither means both fail to reject.

Test the Reverse

Does Y help forecast X?

$$\text{reverse } Y \rightarrow X : F = \frac{(60-58.5)/2}{58.5/95} = \mathbf{1.22} < 3.09 \Rightarrow \text{not significant} \Rightarrow X \rightarrow Y \text{ only}$$

Forecasting the sector from its own past, adding the stock's lags barely lowers the RSS - the reverse F is just 1.22, well below 3.09. So the stock does not Granger-cause the sector: the relationship runs one way, X to Y.

Classify, Then Beware

One-directional - but not causal proof

The classification

- X to Y: significant ($F = 7.39$).
- Y to X: not significant ($F = 1.22$).
- Outcome: X Granger-causes Y only.

The caveat

- A hidden common driver could fake this.
- Or the stock anticipates the sector's news.
- Prediction is not proof of mechanism.

Placing the Pair

The four-outcome grid

Reading the grid

- Columns: does X Granger-

X Granger-causes Y?
Case B: our pair is X → Y only (F 7.39 yes, reverse 1.22 no)

Y → X?	YES	X → Y only	both (feedback)
	NO	neither	Y → X only
	YES	NO (rev)	

- So it lands in X to Y only (green).

Why the Name Misleads

And why it matters in markets

Three ways it isn't cause

- A common factor drives both series.
- One series anticipates the other's news.
- The lead-lag reverses out of sample.

The efficient-markets lens

- Strong return predictability is rare.
- If it were easy, it would be arbitrated away.
- Findings are often fragile - treat with care.

The Granger Routine

Five steps, one honest conclusion

The test is a fixed routine. Fit Y on its own lags, add X's lags, and take the drop in residual error as an F. Compare to the critical value, then test the reverse to fix the direction. What you get is a precise forecasting statement - X's past helps predict Y - not a claim about mechanism. Used with that discipline it's a sharp tool

The test (Case A)

- RSS drop 52 to 45, $F = 7.39$.
- Clears the 3.09 bar.
- X Granger-causes Y.

The direction (Case B)

- Reverse F only 1.22.
- One-directional: X to Y.
- Prediction, not proven cause.

Reproduce the Test Both Ways

Direct Claude Code to check your arithmetic

GOAL Run the Granger F-test in both directions and classify the pair.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - X to Y:  
  * with RSS_r=52, RSS_u=45, T=100, k=5, q=2,  
    compute the F-statistic and the 5% critical value;  
  * decide whether X Granger-causes Y.  
CASE B - reverse and classify:  
  * with RSS_r=60, RSS_u=58.5, compute the reverse F;  
  * classify the pair among the four outcomes.  
Optional: use statsmodels grangercausalitytests on a  
simulated pair and confirm the direction.
```

Check against the slides

- $F(X \rightarrow Y) = 7.39 > 3.09$.
- $F(Y \rightarrow X) = 1.22 < 3.09$.
- critical $F(2,95) = 3.09$.
- outcome: $X \rightarrow Y$ only.

Deliverable: part2_practice.py whose printed numbers match both F-tests and the classification.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the Granger F-test compare?

**Restricted vs
unrestricted forecast
error**

Q2 How do you find the direction of causality?

**Test both ways; each
F is significant or not**

Q3 Why isn't Granger causality real causality?

**Prediction can come
from a confounder or
anticipation**

All three come from the F-test, the reverse test, and the causality caveat.

Three Pitfalls to Avoid

Where Granger causality trips people

Common mistakes

- "Granger causality proves X drives Y."
- "Test only one direction and stop."
- "A significant result will hold out of sample."

The reality

- It's a forecasting claim, not a mechanism.
- Always test both ways to fix the direction.
- Predictability in markets is often fragile.

Two Calculations to Remember

Part 2 applications in one screen

1

The F-test measures the forecast gain

Adding the sector's lags cut the stock's residual error from 52 to 45; scaled into an F-statistic that reached 7.39, well past the 3.09 cutoff. The sector's past genuinely improves the stock's forecast - X Granger-causes Y.

2

Test both ways, then stay humble

The reverse F was only 1.22, so the stock does not help forecast the sector: the relationship is one-directional, X to Y. But a common driver or anticipation could produce the same pattern - it's prediction, not proven cause.

3

A sharp tool, used with discipline

Fit restricted, add lags, take the error drop as an F, and check both directions. The result is an honest forecasting statement - which in efficient markets is rare and often fragile.

Next — Part 3 practice: impulse-response functions and variance decomposition, tracing shocks through the system.